

MySQL vs MongoDB vs Cassandra - Interview Comparison

Aspect	MySQL	MongoDB	Cassandra
DB Type	Relational (RDBMS)	Document DB	Wide-column NoSQL
Data Model	Tables & Rows	JSON/BSON Documents	Tables with Partition Keys
Schema	Fixed	Flexible (schema-less)	Fixed (evolvable)
Primary Key	Optional (recommended)	Mandatory (_id)	Mandatory
Secondary Index	Excellent	Excellent	Limited (avoid at scale)
Joins	Yes	No	No
Transactions	Full ACID	Multi-document supported	Limited (partition level)
Read Latency	~1–5 ms	~2–10 ms	~2–10 ms
Write Latency	~2–10 ms	~2–10 ms	~1–5 ms
Write Throughput	Medium	High	Very High
Horizontal Scaling	Manual Sharding	Native Sharding	Native (peer-to-peer)
Vertical Scaling	Excellent	Good	Limited
Read Pattern	SQL Queries	Document Queries	Primary Key Lookup
Write Pattern	Random Updates	Document Updates	Append/Upsert Optimized
Best Index	B+ Tree	B+ Tree	Partition Key
Storage	Row-oriented	BSON Documents	SSTables + LSM Tree
Query Language	SQL	Mongo Query	CQL
Aggregations	Excellent	Good	Limited
Analytics	Good	Fair	Fair
Full-text Search	Limited	Basic	No
Consistency	Strong	Strong/Eventual	Tunable
Replication	Primary-Replica	Replica Set	Peer-to-Peer
Failover	Automatic	Automatic	Automatic
Billions of Rows	With Sharding	Yes	Designed for it
Typical Use	Banking, Orders	Profiles, Catalogs	Time-series, Logs, Messaging, IoT

Internal Storage

MySQL: InnoDB, row storage, B+ Tree indexes.

MongoDB: WiredTiger, BSON documents, B+ Tree indexes.

Cassandra: LSM Tree, SSTables, Bloom Filters, Partition Index.

Strengths

MySQL: ACID transactions, SQL, complex joins, mature optimizer.

MongoDB: Flexible schema, rich indexing, easy object mapping.

Cassandra: Massive horizontal scaling, millions of writes/sec, excellent key-value lookups, no master bottleneck.

Weaknesses

MySQL: Hard to shard, write scaling is difficult.

MongoDB: Limited joins, multi-document transactions are expensive.

Cassandra: No joins, query-driven schema design, secondary indexes don't scale well.

When to Choose

Banking & Orders → MySQL

User Profiles & Product Catalog → MongoDB

TinyURL, WhatsApp Messages, Metrics, IoT → Cassandra

Interview Rule of Thumb

Need joins + transactions + SQL → MySQL

Need flexible JSON documents → MongoDB

Need billions of rows + millions of writes/sec + primary-key lookups → Cassandra